STRATEGIC APPLICATION OF ARTIFICIAL INTELLIGENCE IN CAD

A Project Report Submitted to
Rajiv Gandhi Proudyogiki Vishwavidyalaya, Bhopal
towards the partial fulfillment of the degree of Bachelor of Technology
in Mechanical Engineering

2026 - 2027

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SHRI G.S. INSTITUTE OF TECHNOLOGY AND SCIENCE, INDORE (M.P.)

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A Govt. Aided Autonomous Institute, Affiliated to RGPV, Bhopal

DEPARTMENT OF MECHANICAL ENGINEERING

2026 - 2027

RECOMMENDATION

We are pleased to recommend that the project work entitled *Strategic Application of Artificial Intelligence in CAD* submitted by Ankur Dodke, Ansh Yadav, Atishay Jain, Bhavya Shrivastava, Dharmendra Gupta may be accepted in partial fulfillment of the degree of Bachelor of Technology in Mechanical Engineering of Rajiv Gandhi Proudyogiki Vishwavidyalaya, Bhopal (M.P.) during the session 2026 - 2027.

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Professor and Head
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DEPARTMENT OF MECHANICAL ENGINEERING

2026 - 2027

CERTIFICATE

This is to certify that the Project entitled *Strategic Application of Artificial Intelligence in CAD* submitted by Ankur Dodke, Ansh Yadav, Atishay Jain, Bhavya Shrivastava, Dharmendra Gupta is accepted in partial fulfillment of the degree of Bachelor of Technology in Mechanical Engineering of Rajiv Gandhi Proudyogiki Vishwavidyalaya, Bhopal during the session 2026 - 2027.

Internal Examiner	**External Examiner**
Date:	**Date:** .....

DECLARATION

We, the undersigned students of the final year Bachelor of Technology course in Mechanical Engineering, Department of Mechanical Engineering, SHRI G.S. INSTITUTE OF TECHNOLOGY AND SCIENCE, INDORE (M.P.), declare that the project entitled *Strategic Application of Artificial Intelligence in CAD* is our own work conducted under the supervision of Dr. B. R. Rawal, Professor and Head, Department of Mechanical Engineering, Shri G. S. Institute of Technology and Science, Indore.

We affirm that this work is entirely original and has not been copied from any source. All sources consulted during the work have been duly acknowledged and cited in the references. Every design formula and empirical proportion used in the software developed for this project is attributed in the text to the published source from which it is taken, and every instance in which the software departs from a cited source is stated explicitly in Chapter 7 together with the reason for the departure.

We further declare that, to the best of our knowledge, this project work does not contain any part of any work which has been submitted for the award of any degree either in this University or in any other University or from any website without proper citation.

Signature of the candidate Date	Signature of the candidate Date	Signature of the candidate Date	Signature of the candidate Date	Signature of the candidate Date
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We thank the faculty of the Department of Mechanical Engineering for their observations during the review presentations, which repeatedly sharpened the scope of the work, and the Institute for providing the facilities and the academic environment in which the work was carried out.

We acknowledge the National Programme on Technology Enhanced Learning and the Indian Institute of Technology Kharagpur, whose openly published course material on the design of machine elements is the primary reference for this work, and the Internet Archive, through which the out of copyright texts by Unwin and by Leutwiler were obtained.

Finally, we thank our families and our classmates for their support during the course of this project.

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ABSTRACT

Designing a standard machine joint is a repetitive calculation. For a knuckle joint the designer computes the rod diameter from direct tension, sizes the pin separately for double shear and for bending, adopts the larger of the two requirements, derives the remaining proportions from the rod diameter, and then verifies nine distinct failure modes against allowable stresses. A change in the applied load invalidates every one of those results and requires the entire sequence again. The drafting that follows is mechanical but unavoidable. Computer aided design software removes the labour of drawing but not the labour of calculation: a modeller regenerates whatever dimensions it is given and will not object if a pin is too slender to carry the bending moment placed upon it.

This project presents a software system that performs the complete sequence automatically from a requirement written in ordinary English. The user enters a sentence such as *knuckle joint for 50 kN*. A language model extracts the parameters from that sentence. All subsequent work is deterministic: the joint is sized from standard design formulas, each computed diameter is rounded up to the next preferred size and each derived thickness to the next whole millimetre, every applicable failure mode is verified against the allowable stress, any dimension that fails is increased until it passes, every component is generated as an exact solid model, and a first angle A3 drawing sheet is produced with dimension lines and a title block.

Six joints are implemented: the knuckle joint, the socket and spigot cotter joint, the sleeve and cotter joint, the gib and cotter joint, the hex bolted joint and the protected type rigid flange coupling. Together they apply 44 distinct failure checks. Every proportion and every check was compared against published design literature, and the comparison identified three errors in the sources themselves, including a worked example whose own answer exceeds its own allowable stress by 23 percent. The implementation is covered by 301 automated tests and is deployed as a web application requiring no installation.

The central design decision is that the language model reads the requirement only. It selects no dimension and performs no check. Interpreting a phrase is a problem of language; selecting a pin diameter is a problem of engineering with a single correct answer, and it remains with deterministic code whose every step can be traced to a printed page.

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LIST OF SYMBOLS

Symbol	Meaning	Unit
P	Applied axial load	N
T	Transmitted torque	N mm
D	Rod diameter	mm
d	Pin or spigot diameter	mm
d_2	Outer diameter of the eye or socket	mm
t	Thickness of the eye, or of the cotter	mm
t_1	Thickness of each fork plate	mm
b	Breadth of the cotter and gib	mm
σ_t	Allowable tensile stress	MPa
σ_b	Allowable bending stress	MPa
σ_c	Allowable crushing stress	MPa
τ	Allowable shear stress	MPa

LIST OF ABBREVIATIONS

Abbreviation	Expansion
CAD	Computer Aided Design
STEP	Standard for the Exchange of Product data
DXF	Drawing Exchange Format
FEA	Finite Element Analysis
NPTEL	National Programme on Technology Enhanced Learning
OCCT	Open CASCADE Technology
BIS	Bureau of Indian Standards
MPa	Megapascal, newton per square millimetre

CHAPTER 1

INTRODUCTION

1.1 Background

A machine joint transmits load between two members. The standard joints taught in machine design, among them the knuckle joint, the cotter joint and the shaft coupling, are not designed by judgement. Each is sized by a fixed sequence of formulas and then verified against a fixed list of failure modes. The sequence is well established and appears in substantially the same form in every standard text on the subject.

Because the sequence is fixed, it is repetitive. A change in the applied load invalidates every dimension and requires the whole calculation to be performed again. In academic practice this repetition consumes the majority of the time spent on a design problem, while the engineering judgement involved, namely the choice of allowable stresses and the interpretation of the result, occupies very little of it. The drafting that follows the calculation is similarly mechanical.

Computer aided design software removes the labour of drawing but not the labour of calculation. A parametric modeller will faithfully regenerate a model when its driving parameters change, and will not object if the pin of a knuckle joint is too slender to carry the bending moment placed upon it. Determining the parameters remains the responsibility of the designer and remains manual.

1.2 Motivation

The motivation for this work is the gap between stating a requirement and obtaining a verified design. At present a designer must translate a requirement such as *a knuckle joint to carry 50 kN* into a sequence of formulas by hand, evaluate them, check the result against several failure modes, iterate wherever a check fails, and only then begin modelling and drafting.

A system that accepts the requirement as written and returns a verified design, complete with solid models and a drawing sheet, would remove the repetition without removing the engineering. The difficulty is that interpreting a written requirement is not a calculation, while the calculation itself must not be delegated to a mechanism that cannot be checked.

1.3 Objectives

The objectives of the present work are as follows.

1. To develop a system that accepts a design requirement stated in ordinary English and extracts from it the joint type, the applied load and the allowable stresses.

2. To size the joint using established design formulas and to verify every applicable failure mode against the allowable stress, iterating wherever a check is not satisfied.
3. To generate exact solid models of every component and of the assembly in a neutral exchange format that opens in commercial CAD software.
4. To generate a first angle projection drawing sheet carrying dimensions and a title block in conformity with Indian drawing practice.
5. To verify every proportion and every failure check used by the system against published design literature, and to record any disagreement with a source explicitly.
6. To make the system usable through a web browser without any installation.

1.4 Scope of the Work

The scope of the present work is limited to six standard joints under static loading. Fatigue loading, stress concentration factors, tolerances and fits, and the selection of manufacturing processes are outside the scope. The finite element facility included in the system performs a single linear static analysis on one component and is intended as an independent cross check on the formula rather than as a substitute for a complete analysis.

1.5 Organisation of the Report

Chapter 2 reviews the design literature consulted and the existing approaches to design automation. Chapter 3 identifies the gap and formulates the problem. Chapter 4 describes the joints selected as case studies and the data collected for them. Chapter 5 sets out the analysis and design methodology, including the division of responsibility between the language model and the deterministic calculation. Chapter 6 describes the implementation. Chapter 7 presents the results, the verification against published sources and the discrepancies found in those sources. Chapter 8 concludes and identifies the scope for further work.

CHAPTER 2

LITERATURE REVIEW

2.1 Design of Cotter and Knuckle Joints

The standard text for the course, Bhandari's Design of Machine Elements, treats the cotter joint and the knuckle joint in sections 4.9 to 4.12. It gives the strength equations for every failure mode, the empirical proportions of each joint, and complete worked examples: Example 4.2 and 4.4 for the cotter joint and Example 4.5 for the knuckle joint. The software follows Bhandari's proportions for the knuckle joint, in which the outside diameter of the eye and the pin head are taken from the pin diameter ($d_0 = 2d$, $1.5d$) and the rod is enlarged to $1.1D$ at the eye, and his procedure for the cotter joint, in which the spigot and the socket are sized from their own tension equations. Those worked examples are the cases against which the software is tested in Chapter 7.

A second reference is the course material on the Design of Machine Elements published by the Indian Institute of Technology Kharagpur through the National Programme on Technology Enhanced Learning. Module 4 of that course treats pin joints, keys and threaded fasteners. Lesson 2 gives the complete design procedure for the socket and spigot cotter joint and for the knuckle joint, including the empirical proportions expressed as multiples of the rod diameter and the full list of failure modes to be verified.

Two further texts were used, both out of copyright and obtained from the Internet Archive. Unwin, in *The Elements of Machine Design* of 1899, treats the gib and cotter joint on page 179 and gives the division of the slot breadth between the cotter and the gib, the thickness of both as a fraction of that breadth, and the permissible range of taper. Leutwiler, in *Elements of Machine Design* of 1917, gives in Article 98 a list of ten distinct ways in which a cotter joint may fail, together with the expression governing each of them.

The availability of two independent treatments of the cotter joint, published eighteen years apart, proved significant to this work. Where the two sources agree, the expression adopted here may be regarded as settled. Where they disagree, the disagreement is itself a finding, and is reported in Section 7.4.

2.2 Design of Shaft Couplings

Leutwiler, in Article 275, presents the analysis of the flange coupling as three groups of failure: failure of the key by shear or by crushing, failure of the coupling bolts by shear or by crushing, and shearing of the flange at the hub. The expressions given there for the bolt diameter are adopted in the present work.

The same article records that no uniform set of proportions for flange couplings had been proposed for adoption as a standard, and that the published proportion tables

represent the practice of individual manufacturers. That observation directly limits what may be claimed about the proportions used in the present work, and the limitation is stated in Section 7.3 rather than passed over. The two key equations, shear across the whole key and bearing across half the key thickness, are taken from Module 4 Lesson 1 of the same NPTEL course.

2.3 Standards Consulted

Indian Standards were consulted for drawing practice and for the fasteners and keys used in the joints implemented. SP 46 governs first angle projection and the title block. IS 2048 covers parallel keys and keyways, IS 1363 and IS 1364 cover hexagon head bolts, screws and nuts, and IS 2693 covers cast iron flange couplings.

2.4 Existing Approaches to Design Automation

Parametric modelling is standard in commercial CAD software. A model driven by a table of parameters regenerates when those parameters change, which removes the drafting labour associated with a change of size. It does not, however, determine the parameters. The designer must still compute them, and the software will regenerate an unsafe model as readily as a safe one.

Recent work applying large language models to computer aided design has largely taken the form of generating modelling code directly from a description. Such systems produce geometry quickly, but the dimensions they produce cannot be traced to a design calculation, and there is no mechanism by which a failure mode is verified. For an academic or a professional design task, geometry that cannot be justified is of limited value.

A third group of approaches embeds engineering knowledge in expert systems and design handbooks implemented as software. These are reliable within the range for which they were written, but the calculation is typically hidden from the user, so a result cannot be checked against a source without access to the implementation.

2.5 Comparison with Existing Tools

Table 2.1 compares the present work with seven existing tools and research systems. The entries are taken from each vendor's website, or from the published paper in the case of Text2CAD, as stated on 29 October 2026. They record what each source claims, not what was found by testing the tool, and an entry marked as not stated means only that the source does not state the feature.

Table 2.1: Comparison of the present work with existing tools, as stated on the vendors' websites on 29 October 2026

Tool	Natural language input	Sizing traceable to formulas	Failure mode checks	Standard size rounding	2D drawing sheet	STEP export
BuildCAD AI	Yes	Not stated	Not stated	Not stated	Not stated	Yes
Zoo Text-to-CAD	Yes	Not stated	Not stated	Not stated	Not stated	Yes
CADAM (Adam)	Yes	Not stated	Not stated	Not stated	DXF only	No
Leo AI	Yes	Cited calculations	Not stated	Not stated	Not stated	Not stated
MITCalc	No	Yes	Not stated	Not stated	DXF insert	Not stated
KISSsoft	No	Yes	Not stated	Not stated	Not stated	Not stated
Text2CAD (NeurIPS 2024)	Yes	No	Not stated	Not stated	Not stated	Not stated
Present work	Yes	Yes	Yes	Yes	Yes	Yes

Two groups are visible. The tools that accept natural language, namely BuildCAD AI, Zoo Text-to-CAD, CADAM and Text2CAD, generate geometry, and none of them states that a dimension is sized from a design formula or verified against a failure mode. The tools that do carry out machine element calculation, namely MITCalc and KISSsoft, work from form input rather than from a written requirement. Leo AI is the nearest to the present work in that it cites its calculations, but its output formats are not stated. No source examined states the combination of natural language input, traceable sizing, failure checks, rounding to standard sizes and a drawing sheet.

CHAPTER 3

GAP IDENTIFICATION AND PROBLEM FORMULATION

3.1 Gap Identified

From the review presented in Chapter 2, the gap lies neither in modelling nor in language interpretation, both of which are mature, but in the connection between them. What is absent is a system in which a written requirement is interpreted by a language model, the resulting parameters drive a deterministic design calculation taken from published literature, and the calculation itself remains open to inspection by the user. Table 2.1 shows that none of the tools compared in Section 2.5 states all of these capabilities together.

Three specific deficiencies follow from this. First, parametric CAD regenerates geometry but verifies no failure mode. Second, systems that generate modelling code from a description produce dimensions that cannot be attributed to any formula. Third, where design calculation is automated, the working is generally hidden, so the result must be trusted rather than checked.

3.2 Problem Formulation

The problem addressed in this work is stated as follows. Given a design requirement expressed in ordinary English, produce a complete design of the specified machine joint such that every dimension is obtained from a published design formula or from a named failure check, every applicable failure mode is verified against its allowable stress, the geometry is generated in a neutral exchange format, a first angle drawing sheet is produced, and the entire working is reported to the user in a form that permits independent verification.

3.3 Conditions the Solution Must Satisfy

1. No dimension may be produced without an attributable basis, which is either a design formula with a cited source or an explicitly named failure check.
2. The language model may interpret the requirement but may not select any dimension and may not perform any check.
3. Any numerical value returned by the language model that does not appear in the original requirement must be discarded, so that an invented figure cannot enter the calculation.
4. Every applicable failure mode must be evaluated on every run and its arithmetic reported, not merely its outcome.
5. Where the empirical proportions used have no supporting source, the system must state that fact rather than imply a source it does not have.

CHAPTER 4

ABOUT THE CASE STUDY AND DATA COLLECTION

4.1 Selection of Case Studies

Six joints were selected. They were chosen because each is a standard joint with a published design procedure, because between them they exercise every common failure mode (tension, double shear, crushing, bending and torsion), and because they differ sufficiently in geometry to test whether the system generalises. Table 4.1 lists them.

Table 4.1: Joints selected as case studies, with the duty adopted for each

Joint	Components	Dimensions	Failure checks	Duty adopted
Knuckle joint	5	13	9	Load P = 50,000 N
Socket and spigot cotter joint	3	14	10	Load P = 35,000 N
Sleeve and cotter joint	5	12	8	Load P = 50,000 N
Gib and cotter joint	4	20	8	Load P = 50,000 N
Hex bolt, nut and washer	3	11	2	Load P = 20,000 N
Protected type rigid flange coupling	12	15	7	Torque T = 500,000 N mm (500 N m)

The flange coupling differs from the remaining five in that it is sized from a transmitted torque rather than from an axial load, and in that the number of bolts varies with the shaft diameter. It was included specifically to confirm that the system is not limited to axially loaded joints.

4.2 Data Collected

For each joint, three categories of data were collected from the sources listed in Chapter 2: the empirical proportions expressed as multiples of the governing dimension, the complete list of failure modes with the expression governing each, and the worked numerical example given in the source, which was later used to verify the implementation.

The allowable stresses and the preferred size series adopted are given in Table 4.2. The preferred size series is significant: the source material directs that a computed diameter be rounded up to the next standard size rather than used as computed, and the system follows that direction.

Table 4.2: Allowable stresses adopted, and the sizes to which computed diameters are rounded up

Quantity	Value adopted	Source
Allowable tensile stress	75 MPa	working value used in the source material
Allowable shear stress	60 MPa	working value used in the source material
Allowable crushing stress	150 MPa	working value used in the source material
Design factor	1.25	sized for 1.25 times the load, reported at the load
Preferred diameters	6 to 22 in steps of 2 mm, 25 to 60 in steps of 5 mm, thereafter 10 mm	standard shaft sizes, NPTEL Module 4 Lesson 2

4.3 Basis for Verification

The worked examples given in the sources form the basis on which the implementation was verified. For the knuckle joint and the cotter joint, the source provides a complete numerical example with the load, the allowable stresses and every resulting dimension. Those examples were reproduced by the system and compared value by value. The outcome of that comparison, including the instances in which the system and the source disagree, is reported in Chapter 7.

CHAPTER 5

ANALYSIS AND DESIGN METHODOLOGY

5.1 Overall Approach

The system is organised as five stages executed in sequence. The written requirement is interpreted into parameters; the joint is sized; every failure mode is verified; the geometry is generated; and the drawing sheet is produced. Each stage passes a definite result to the next, and the intermediate result of every stage is reported to the user.

5.2 Role of the Language Model

The language model is used at the first stage only. It receives the written requirement and returns the joint type, the applied load in newtons, and any allowable stresses stated. It selects no dimension, applies no proportion and performs no check. Table 5.1 sets out the division of responsibility.

Table 5.1: Division of responsibility between the language model and the deterministic calculation

Stage	Performed by	Verifiable against
Interpretation of the requirement	Language model	the requirement as written
Sizing of every dimension	Deterministic calculation	cited design formula
Verification of failure modes	Deterministic calculation	cited design formula
Generation of solid geometry	Modelling kernel	the computed dimensions
Generation of the drawing sheet	FreeCAD	the assembled geometry

This division is deliberate and is the principal design decision of the project. Determining that the phrase *50 kN* denotes a load of 50,000 N, and that *socket and spigot* names the cotter joint, is a problem of language interpretation, at which such a model is effective. Determining the diameter of the pin is a problem of engineering with a single correct answer obtainable from a formula, and it is therefore performed by deterministic code whose every step is traceable to a published page.

A safeguard is applied to the output of the model. Any numerical value it returns that does not appear in the original requirement is discarded and the user is informed. This prevents a plausible but invented figure from entering the calculation. Where no language model is available, the system falls back to pattern matching rules and reports which interpreter was used.

5.3 Sizing Procedure

The governing dimension is computed first from the primary failure mode. For a knuckle joint this is the rod diameter obtained from direct tension, given by Equation (5.1).

$$D = \sqrt{(4P / (\pi \sigma_t))} \qquad (5.1)$$

The computed value is then rounded up to the next standard preferred size rather than used as computed, in accordance with the practice given in the source material.

Where a component has more than one independent failure mode, as the pin of a knuckle joint has in both double shear and bending, that component is sized separately from each applicable expression, given by Equations (5.2) and (5.3), and the larger requirement is adopted.

$$d = \sqrt{(2P / (\pi \tau))} \qquad (5.2)$$

$$d = [16P (t_1/3 + t/4) / (\pi \sigma_b)]^{1/3} \qquad (5.3)$$

The remaining dimensions follow from the empirical proportions given in the source, expressed as multiples of the governing dimension.

5.4 Design Factor

A design factor of 1.25 is applied. The joint is sized for 1.25 times the stated load, and the resulting stresses are then reported at the actual load. The consequence is that the governing check reports approximately 75 to 80 percent of the allowable stress rather than 100 percent, so that margin remains in hand. Sizing exactly to the allowable produces a design that satisfies every check on paper while retaining nothing for material variation or for overload.

5.5 Verification of Failure Modes

Every applicable failure mode is evaluated and compared with the corresponding allowable stress. Where a check is not satisfied, the dimension responsible is increased by one millimetre and the entire check set is evaluated again. The process repeats until all checks are satisfied. The dimension that was increased, and the check that caused the increase, are both reported to the user.

5.6 Generation of Geometry and Drawings

Each component is generated as an exact solid using a parametric modelling kernel and is written in the STEP neutral exchange format, which opens in Siemens NX, SolidWorks, FreeCAD, Onshape and Fusion. The assembly is written as a further STEP file with each component in its assembled position.

The drawing sheet is produced by driving FreeCAD without its graphical interface. The assembled geometry is projected to obtain the front view, the left side view and the plan, arranged in first angle projection in accordance with Indian

practice, together with an isometric view, dimension lines and a title block. The same views are also written in DXF format.

5.7 Finite Element Cross Check

An optional finite element analysis is provided as an independent check on the calculated stress. One component is meshed and solved as a linear static problem using the CalculiX solver, and the resulting peak stress is compared with the value obtained from the formula. Where the computed peak exceeds the calculated value by more than half, the analysis is repeated on a mesh of half the element size. If the peak then moves by more than five percent the result is reported as not converged, since a peak that continues to change with mesh refinement indicates a stress concentration at a sharp corner or at a constrained face rather than a stress to which a component may be designed.

5.8 Material Selection

Material selection is a separate step between interpreting the requirement and building the joint. The software holds six materials with their Indian standard designations: the plain carbon steels 30C8, 40C8 and 45C8 to IS 1570, the alloy steel 40Cr1, grey cast iron FG 260 to IS 210 and the aluminium alloy 64430 to IS 733. For each it records working allowable stresses in tension, shear and crushing, the density, and an indicative rate per kilogram.

When a requirement has been read, the joint is designed completely in every material: the governing dimension is sized from that material's allowable stresses, every failure check is evaluated, and the mass is obtained from the volume of the generated solids. The user is shown which materials pass, the size each requires, the mass and the material cost, with the lightest and the cheapest passing material marked. The material the user selects then supplies the allowable stresses for every check in the design. Table 5.2 gives the result for the worked example of Section 7.2.

Table 5.2: Knuckle joint for 50 kN designed in each material offered, as computed by the software

Material	Grade	σ_t, MPa	τ, MPa	σ_c, MPa	D, mm	Mass, kg	Cost, Rs
30C8 plain carbon steel	IS 1570 Part 2, 30C8	75	60	150	35	11.36	795
40C8 plain carbon steel	IS 1570 Part 2, 40C8	95	70	180	30	7.63	611
45C8 plain carbon steel	IS 1570 Part 2, 45C8	110	80	200	30	7.63	687
40Cr1 alloy steel (cheapest)	IS 1570 Part 4, 40Cr1	160	110	280	25	4.05	607
Grey cast iron FG 260	IS 210, FG 260	40	35	250	45	22.43	1,346
Aluminium alloy 64430 (6061) (lightest)	IS 733, 64430	80	55	140	32	3.39	951

The comparison makes the trade off explicit. A stronger steel reduces the rod diameter and the mass, while cast iron, weak in tension, requires the largest section. The allowable stresses and the rates in the table are working values; before they are used for a real component they are to be confirmed against a design data book and a supplier's quotation, and the software states this on every run.

CHAPTER 6

IMPLEMENTATION

6.1 System Architecture

The design rules for each joint are held as data rather than as code. Each joint is described by a file recording its empirical proportions, the source of each proportion, the default allowable stresses and the preferred size series. The calculation code reads these files. Two consequences follow. The source of any dimension can be reported to the user at the moment it is computed, and a correction to a proportion is a change to data rather than to code.

6.2 Software Tools

Table 6.1 lists the tools used. All of them other than the language model are open source.

Table 6.1: Software tools used in the implementation

Function	Tool	Remark
Solid modelling	CadQuery 2.8 on the OCCT kernel	open source, scripted parametric modelling
Drawing sheet and DXF	FreeCAD, without its interface	projection, dimensioning and title block
Finite element analysis	CalculiX with gmsh	linear static, single component
Interpretation of the requirement	Claude language model	parameter extraction only
Application	Python standard library	no external web framework
Deployment	container on a public host	no installation required of the user

6.3 Implementation of the Joints

Each joint is implemented as a module exposing the same interface: a sizing function, a list of failure checks, a geometry builder and a set of dimension specifications for the drawing sheet. Adding a joint therefore requires no change to the surrounding system, which is what allowed the gib and cotter joint to be added after the other five were complete.

The gib and cotter joint requires particular comment. Unwin gives the division of the breadth and the thickness of the gib and cotter but gives no proportions for the strap. Rather than adopt a ratio from an unrelated source or invent one, every dimension the source does not supply is sized against a failure check until that check is satisfied. The result is that no dimension of that joint is unsupported: each is either taken from the source or derived from a stated failure mode.

6.4 User Interface

The system is used through a web browser. The user enters the requirement, confirms or corrects the parameters extracted from it, and requests the build. The complete working of the calculation is printed as it proceeds. The results are presented as the three dimensional model, the drawing sheet, the list of dimensions, the list of checks, a view colouring components by utilisation, and the generated files.

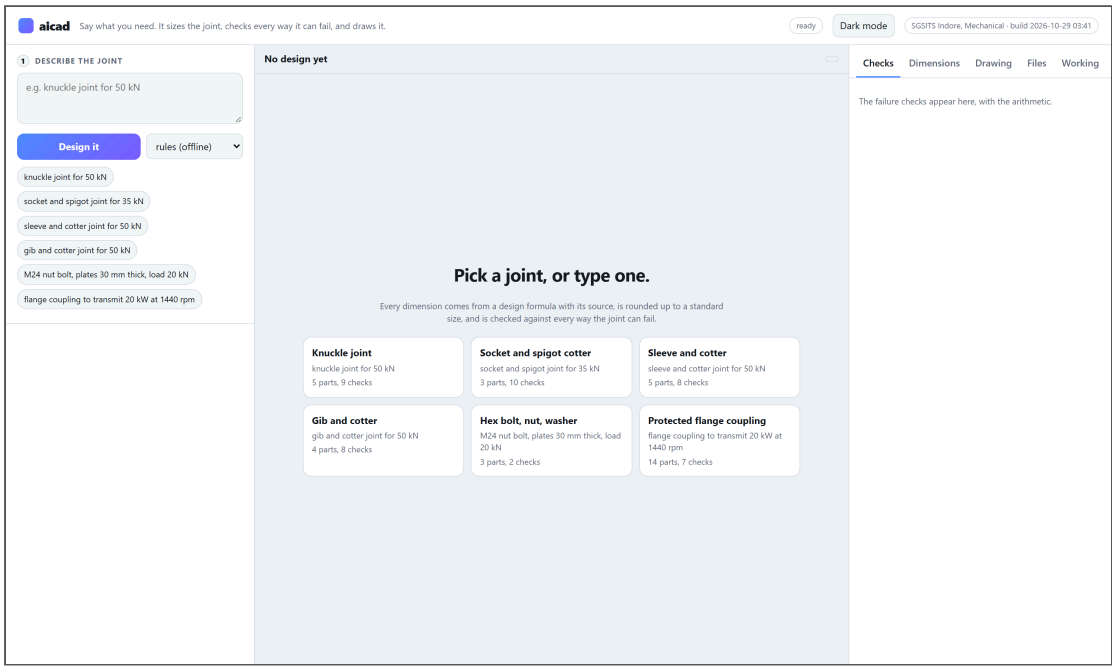


Figure 6.1: Entry screen of the system, showing the requirement field and the example requirements provided

Checks Dimensions Drawing Files **Working**

```
Read as: Knuckle joint, Load P = 50,000 N
  formula rod_tension: 32.57 mm
  formula pin_shear: 25.75 mm
  formula pin_bending: 43.95 mm
  D = 35 mm  <-  rod in tension: D = sqrt(4P / (pi
sigma_t)), rounded up to a preferred size
  d = 45 mm  <-  governed by pin bending
  d2 = 70 mm  <-  d2 = 2.0 x D, the rod diameter (outer
diameter of eye), rounded up to a whole mm [NPTEL IIT KGP
M4L2 fig 4.2.3.1]
  d3 = 55 mm  <-  d3 = 1.5 x D, the rod diameter (pin head
and collar diameter), rounded up to a preferred size
[NPTEL IIT KGP M4L2 fig 4.2.3.1]
  t = 44 mm  <-  t = 1.25 x D, the rod diameter (thickness
of single eye), rounded up to a whole mm [NPTEL IIT KGP
```

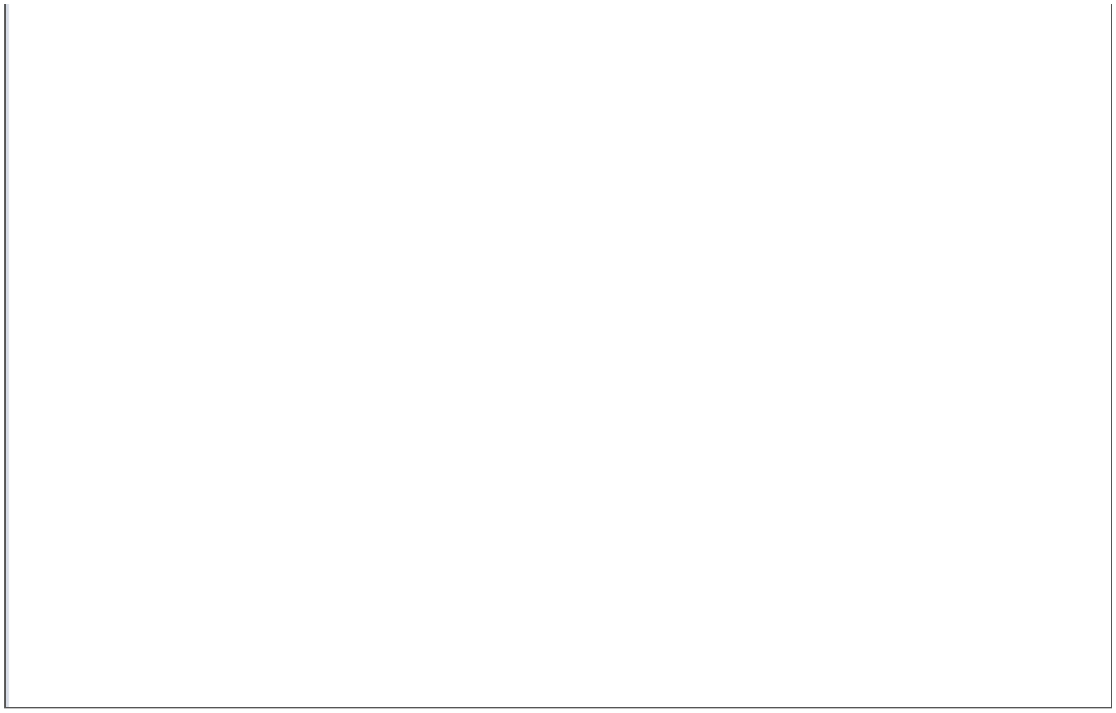


Figure 6.2: Working printed during a single run, showing the values computed before rounding, the rule and source of each proportion, and every check with its arithmetic substituted

6.5 Verification of the Implementation

The implementation is covered by 301 automated tests which are executed on every change. These cover the sizing arithmetic, the generated geometry, the interpretation of requirements, the drawing sheet and the finite element facility. Among them are tests which assert that the discrepancies recorded in Section 7.4 are genuine, so that the system cannot subsequently be altered to reproduce the printed answers without a test failing.

One class of test deserves mention because it detected an error that no dimensional check could have detected. The geometry tests compare the solids of every pair of components and require that no two occupy the same space. That test found a hooked gib modelled 1.25 mm inside the strap, an error invisible in the dimension list because every individual dimension was correct.

6.6 Build Time

The time taken by each stage of a build was measured. Table 6.2 gives the result. The calculation and the solid models take about a second between them, and the drawing sheet produced by FreeCAD accounts for about 33 s of a 35 s build.

Table 6.2: Measured time of each stage of a build

Stage	Time
Sizing and the nine failure checks	under 1 ms
Five solid models	0.9 s
STEP and viewer files	negligible
FreeCAD drawing sheet	about 33 s of the 35 s build

Three changes follow from this measurement. The system now returns the model and the failure checks first and produces the drawing sheet in the background, so that the user can read the result while the sheet is still being drawn. Identical requests are cached, so that a repeated request does not repeat the work. The example joints offered on the entry screen are built in advance.

CHAPTER 7

RESULTS AND DISCUSSION

7.1 Design Results

All six joints size correctly, satisfy every applicable failure check, generate every component as a solid model and produce a drawing sheet. Each design completes in approximately ten seconds. Table 7.1 summarises the results.

Table 7.1: Design results for the six joints at the duties given in Table 4.1

Joint	Components	Checks	Busiest check, percent of allowable	Outcome
Knuckle joint	5	9	79	all satisfied
Socket and spigot cotter joint	3	10	76	all satisfied
Sleeve and cotter joint	5	8	77	all satisfied
Gib and cotter joint	4	8	79	all satisfied
Hex bolt, nut and washer	3	2	71	all satisfied
Protected type rigid flange coupling	12	7	79	all satisfied

The busiest check across the six joints lies between 71 and 79 percent of the allowable stress. That range is the intended consequence of the design factor described in Section 5.4 and confirms that margin is retained in every case.

7.2 Worked Example

The requirement *knuckle joint for 50 kN* is interpreted as the knuckle joint carrying an axial load of 50,000 N with the default allowable stresses. Applying the design factor, the joint is sized for 62,500 N. Table 7.2 gives the three governing calculations.

Table 7.2: Governing calculations for a knuckle joint carrying 50 kN, before rounding

Expression	Computed, mm	Effect
Rod in direct tension	32.57	rod diameter adopted as 35 mm
Pin in double shear	25.75	not governing
Pin in bending	44.85	governs; pin diameter adopted as 45 mm

The requirement arising from bending is nearly 1.7 times that arising from shear. A pin sized from shear alone would satisfy its own check and bend in service. This is the reason the larger of the two requirements is adopted, and the source

material takes the same step. Table 7.3 gives the complete set of dimensions with the rule and the source behind each.

Table 7.3: Dimensions computed for the knuckle joint, with the rule behind each

Symbol	Meaning	Value, mm	Rule and source
D	rod diameter	35	rod in tension: D = sqrt(4P / (pi sigma_t)), rounded up to a preferred size
d	pin diameter	45	governed by pin bending
d2	eye outer diameter	90	d2 = 2.0 x d, the pin diameter (outer diameter of eye and fork, d0 = 2d), rounded up to a standard size [Bhandari, Design of Machine Elements, sec 4.1
d3	pin head and collar diameter	70	d3 = 1.5 x d, the pin diameter (pin head and collar diameter, 1.5d), rounded up to a standard size [Bhandari, Design of Machine Elements, sec 4.12, eq
D1	enlarged end, 1.1D	40	D1 = 1.1 x D, the rod diameter (enlarged rod diameter near the eye and fork, D1 = 1.1D), rounded up to a standard size [Bhandari, Design of Machine El
t	eye thickness	45	t = 1.25 x D, the rod diameter (thickness of eye of rod B, b = 1.25D), rounded up to a standard size [Bhandari, Design of Machine Elements, sec 4.12,
t1	fork plate thickness	30	t1 = 0.75 x D, the rod diameter (thickness of each fork eye, a = 0.75D), rounded up to a standard size [Bhandari, Design of Machine Elements, sec 4.12
t2	pin head thickness	18	t2 = 0.5 x D, the rod diameter (thickness of pin head), rounded up to a whole mm [NPTEL IIT KGP M4L2 fig 4.2.3.1]
collar_t	collar thickness	23	collar_t = 0.5 x d, the pin diameter, rounded up to a whole mm; modelling choice, not a book value
taper_d	taper pin diameter (big end)	9	taper_d = 0.25 x D, the rod diameter (split pin mean diameter), rounded up to a whole mm [NPTEL IIT KGP M4L2]

With all dimensions determined, the nine failure modes are evaluated at the actual load of 50,000 N. The results are given in Table 7.4. Every check is satisfied, and the busiest, shear of the pin in bending, reaches 79 percent of its allowable stress.

Table 7.4: Failure checks for the knuckle joint, evaluated at the actual load of 50 kN

Failure mode	Expression	Stress, MPa	Allowable, MPa	Utilisation, percent	Result
rod tension	`4P / (pi D^2)`	52	75	69	safe
pin double shear	`P / (2 pi/4 d^2)`	15.7	60	26	safe
pin bending	`32M / (pi d^3), M = P/2 (t1/3 + t/4)`	59.4	75	79	safe
eye tension	`P / ((d2 - d) t)`	24.7	75	33	safe
eye shear	`P / ((d2 - d) t)`	24.7	60	41	safe
eye crushing	`P / (d t)`	24.7	150	16	safe
fork tension	`P / ((d2 - d) 2 t1)`	18.5	75	25	safe
fork shear	`P / ((d2 - d) 2 t1)`	18.5	60	31	safe
fork crushing	`P / (d 2 t1)`	18.5	150	12	safe

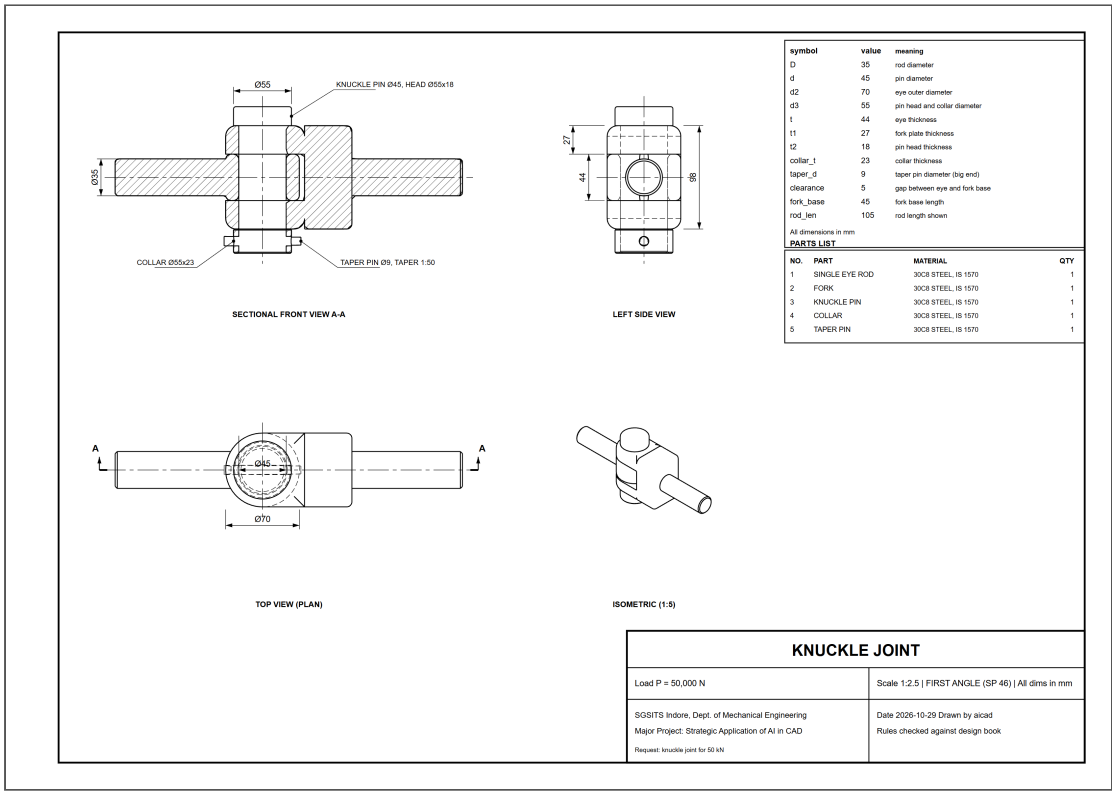


Figure 7.1: First angle A3 drawing sheet generated for the knuckle joint of Section 7.2, at a scale of 1:2.5

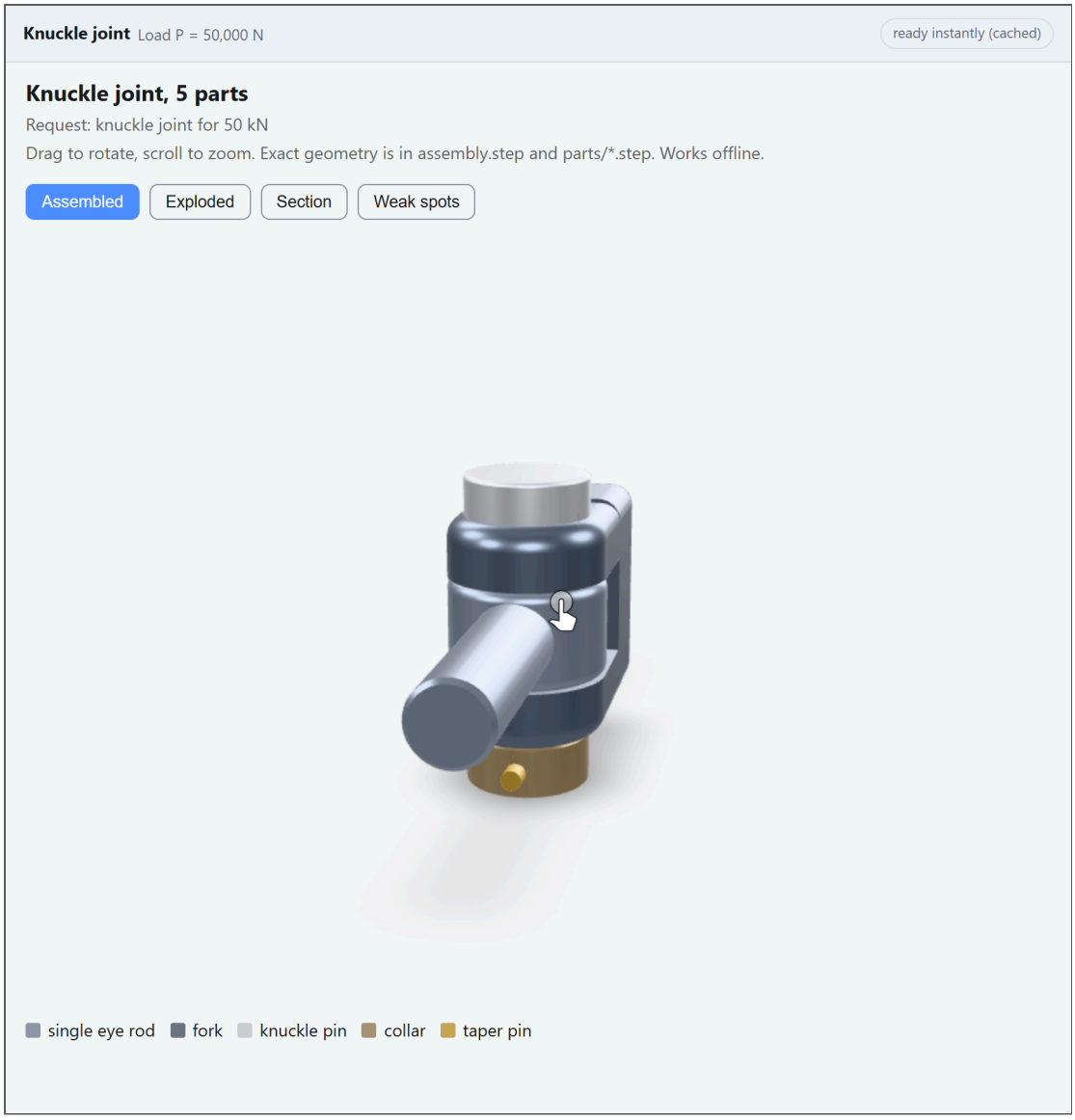


Figure 7.2: Assembled model of the same joint, also available exploded and coloured by utilisation

7.3 Verification Against Published Literature

Every proportion and every failure check was compared against the sources reviewed in Chapter 2. The outcome is given in Table 7.5.

Table 7.5: Verification of each joint against published design literature

Joint	Source consulted	Outcome
Knuckle joint	Bhandari Ex 4.5 and Problem 4.1; NPTEL Module 4 Lesson 2	all seven adopted sizes and six check stresses of Ex 4.5 reproduced exactly; nine checks verified
Socket and spigot cotter joint	Bhandari Ex 4.2; NPTEL Module 4 Lesson 2; Leutwiler Article 98	nine of ten sizes of Ex 4.2 reproduced; ten checks verified against two independent sources
Gib and cotter joint	Unwin, page 179	breadth division, thickness and taper verified
Protected type flange coupling	NPTEL Module 4 Lesson 1; Leutwiler Article 275	all seven checks verified twice; proportions not verifiable
Sleeve and cotter joint	four sources searched, none covers it	checks verified by correspondence; proportions unverified
Hex bolted joint	NPTEL Module 4 Lesson 4	tightening relation verified; size table unverified

Two entries require explanation. For the flange coupling, all seven failure checks are verified against two independent sources, but the empirical proportions are marked unverified. This is not an omission. As recorded in Section 2.2, Leutwiler states that no standard set of proportions was ever agreed and that published tables reflect the practice of individual manufacturers. The proportions used are therefore a convention, and the system reports them as such on every run. For the sleeve and cotter joint, four sources were searched and none treats that joint by name; its checks are verified by direct correspondence with the socket and spigot cotter joint, but its proportions remain unsupported and are reported as such.

The strongest test is the reproduction of complete worked examples. Given the inputs of Bhandari's Example 4.5 (50 kN, 30C8, factor of safety 5), the software adopts exactly the sizes the book adopts: rod 30 mm, enlarged rod 35 mm, eye 40 mm and fork 25 mm thick, pin 40 mm, eye diameter 80 mm and pin head 60 mm, and it computes the same six eye and fork stresses, 31.25 and 25 N/mm^2. Problem 4.1 is reproduced to the printed 14.47 mm and 13.47 mm. For the cotter joint of Example 4.2 nine of the ten adopted sizes agree, including the spigot and socket diameters solved from their tension equations (37.91 and 52.04 mm before rounding). The one difference, the width of the cotter, follows from the book designing the cotter alone with a factor of safety of 4 while the software applies the rods' factor of 6 to every part. Each of these cases is an automated test and a one click case in the web application.

The two texts disagree on one proportion of the knuckle joint. The NPTEL lesson takes the pin head as 1.5 times the rod diameter, which on its own 55 mm pin

leaves a head only 5 mm larger than the pin; Bhandari takes it as 1.5 times the pin diameter. The software follows Bhandari.

7.4 Discrepancies Found in the Sources

The verification exercise identified three errors in the published sources. They are recorded here because the behaviour of the system deliberately differs from the printed material in consequence.

7.4.1 A Worked Example That Fails Its Own Check

The knuckle joint worked problem raises the pin diameter from 40 mm to 55 mm to satisfy bending, but subsequently evaluates the eye using the original 40 mm pin. Evaluated with the pin actually adopted, its own expression for shear of the eye gives 80 MPa against an allowable of 65 MPa, an exceedance of 23 percent. The present system instead enlarges the outer diameter of the eye and reports that it has done so.

7.4.2 An Omitted Failure Mode

The cotter joint procedure in the same source verifies crushing of the socket but does not verify crushing between the cotter and the spigot. Evaluated at the final answer given in that source, the omitted check gives 167 MPa against an allowable of 110 MPa. Leutwiler, in Article 98, lists this mode as one of ten, and its expression is identical in form. The present system retains the check and sizes the cotter thicker in consequence. That an independent text, published eighteen years earlier, lists a check which the lesson omits is what establishes this as an omission rather than a matter of preference.

7.4.3 A Constant Printed Incorrectly

The initial tightening load for a bolted joint is printed as 284*d* kN. The worked example on the same page obtains 56.8 kN for an M20 bolt, which is consistent only with a constant of 2840*d* newtons. The printed relation is therefore in error by a factor of one hundred, and the source refutes itself. The system uses 2840*d* N and reports the tightening load on every bolted design, since at M20 it amounts to 56,800 N before any service load is applied.

7.5 Finite Element Comparison

The finite element facility was first validated against a case with a known answer. Table 7.6 gives the comparison.

Table 7.6: Comparison of the finite element result with the calculated stress

Case	Calculated, MPa	Computed, MPa	Difference, percent
Bar of 20 mm diameter, 10 kN in tension	31.83	32.22	+1.2
Knuckle joint eye rod, 50 kN	52.0	57.9	+11.3

The bar validates the method. The eye rod reads higher because the calculation treats the rod as a plain bar in tension and takes no account of the stress raised at the pin bore, whereas the finite element model does. Both values remain below the allowable stress of 75 MPa.

The convergence rule described in Section 5.7 proved necessary in practice. For one component the computed peak varied between 157 and 191 MPa across a sixteen fold increase in the number of nodes without settling, which indicates a singularity at the constrained face rather than a physical stress. The system reports such a result as not converged and directs the user to the calculated value.

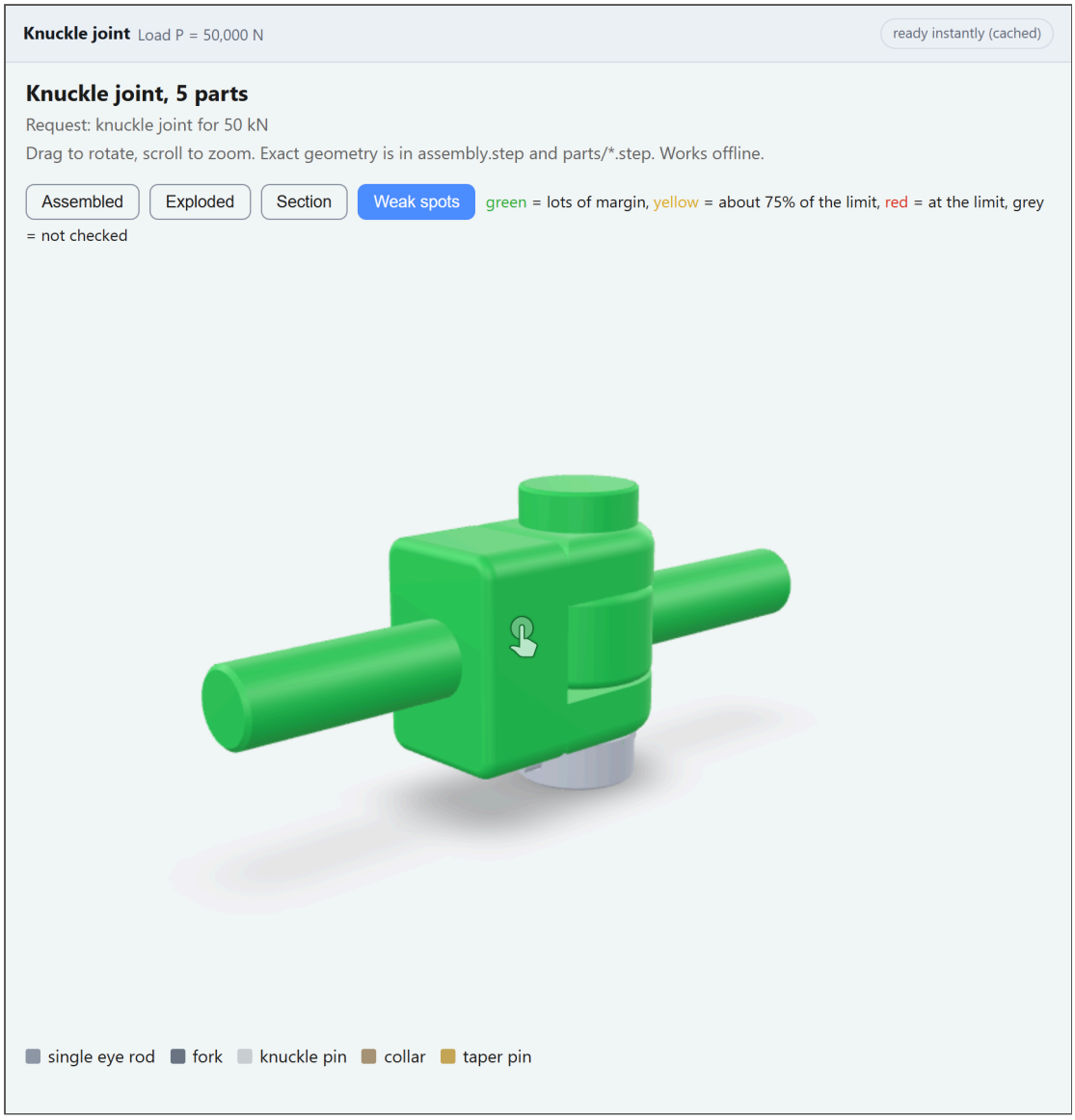


Figure 7.3: Components coloured by utilisation, so that the member nearest its limit is apparent without reading the table

7.6 Discussion

The results support the approach adopted. The division of responsibility set out in Section 5.2 allowed a natural language interface to be provided without any part of the design becoming unverifiable, and the requirement that every check be evaluated and its arithmetic printed is what made the three discrepancies of Section 7.4 visible. None of them was found by inspection of the sources; all three surfaced because a number computed by the system disagreed with a number printed in a book, and the disagreement could be traced.

7.7 Objectives Revisited

Section 1.3 set out six objectives. Each is assessed against the evidence presented above, rather than restated as met without qualification.

1. **Interpreting a plain English requirement.** Met. The language model extracts the joint type and load, as shown in Section 5.2, and any value it returns that cannot be traced back to the original sentence is discarded rather than used, which is the safeguard against a fabricated input rather than a verified calculation.
2. **Sizing and verifying against every failure mode, with iteration.** Met. All six joints satisfy all 44 applicable checks at the stated load, as given in Table 7.1, and the iterative enlargement described in Section 5.5 is what makes the gib and cotter strap, and the knuckle eye of the NPTEL problem, pass where the source proportions alone would not.
3. **Exact solid models in a neutral format that opens in commercial CAD software.** Substantially met. Every exported assembly was inspected directly and confirmed to carry the AUTOMOTIVE_DESIGN (AP214) schema with each component as a separate named solid rather than a single fused body, which is the structure Siemens NX requires to recognise an assembly on import. Opening the file inside NX itself, available only in the departmental laboratory, is the one objective not yet confirmed by direct observation and is recorded as such rather than assumed.
4. **A first angle drawing sheet conforming to Indian practice.** Met for projection and sheet layout, as shown in Figure 7.1. Dimensioning to SP 46 tolerance and surface finish conventions is not implemented and is listed in Section 8.2 rather than claimed here.
5. **Verification against published literature, with disagreements recorded.** Met for four of six joints, and partially met for the remaining two, exactly as tabulated in Table 7.5. The three discrepancies of Section 7.4 are the direct product of this objective rather than a side finding.
6. **Browser use without installation.** Met. The system is deployed and reachable without a local install, though the hosted copy runs the rule based interpreter rather than the language model, a distinction recorded in Section 5.2 rather than left implicit.

Taken together, Objectives 1, 2 and 6 are met without qualification. Objectives 3, 4 and 5 are met in part, and the open point in each is stated above: the import into NX is not yet observed, dimensioning to SP 46 tolerances is not implemented, and two joints remain partly unverified. Reporting that distinction is itself a requirement

of Objective 5, which calls for disagreements and open points to be recorded explicitly rather than smoothed over.

CHAPTER 8

CONCLUSIONS AND SCOPE OF FUTURE WORK

8.1 Conclusions

A system has been developed which accepts a machine joint requirement written in ordinary English and returns a complete design: every dimension computed from published design formulas, every applicable failure mode verified against the allowable stress, every component generated as an exact solid model, and a first angle drawing sheet produced with dimensions and a title block. Six joints are implemented, applying 44 distinct failure checks in total, and the implementation is covered by 301 automated tests.

The objective of keeping the design traceable has been met. Every dimension the system produces can be attributed either to a formula with a cited source or to an explicitly named failure check. The working of each run is printed in full, which allows the result to be verified rather than accepted.

The verification exercise against published literature produced an unanticipated result. Three errors were identified in the sources themselves, one of which is a worked example whose own answer exceeds its own allowable stress by 23 percent. This was found not by inspection but because the system evaluates every check on every design and reports the arithmetic, which makes an omitted or inconsistent check visible immediately.

The division of responsibility adopted here, in which the language model interprets the requirement and deterministic code performs the engineering, is the element of the work that is most generally applicable beyond the six joints implemented.

8.2 Scope for Future Work

1. **Additional joints.** Riveted and welded joints, treated in Module 10 of the source material, follow the same pattern and could be added without change to the architecture.
2. **Fatigue loading.** All sizing at present is static. Introducing an endurance limit and a stress concentration factor would extend the system to cyclically loaded joints.
3. **Tolerances and fits.** The drawing sheet carries nominal dimensions only. Adding fits for the pin and its bore would make the output directly usable for manufacture.
4. **Verification of the remaining proportions.** Three of the six joints have proportions without a supporting source: the sleeve and cotter joint, the flange

coupling and the size table of the hex bolted joint. A design data handbook would settle all three.

5. **Direct integration with commercial CAD.** Output is presently exchanged as STEP. A journal script for Siemens NX would allow the assembly to be constructed within that software rather than imported into it.
6. **Manufacturing outputs.** Material selection and cost estimation are partially implemented and could be extended to produce a bill of materials.

APPENDIX A

SCREENSHOTS OF THE WORKING SYSTEM

The system described in this report is deployed and can be used without installation at **aicad-vercel.vercel.app**. Scan the code or enter the address in a browser; the screenshots below were captured from it.

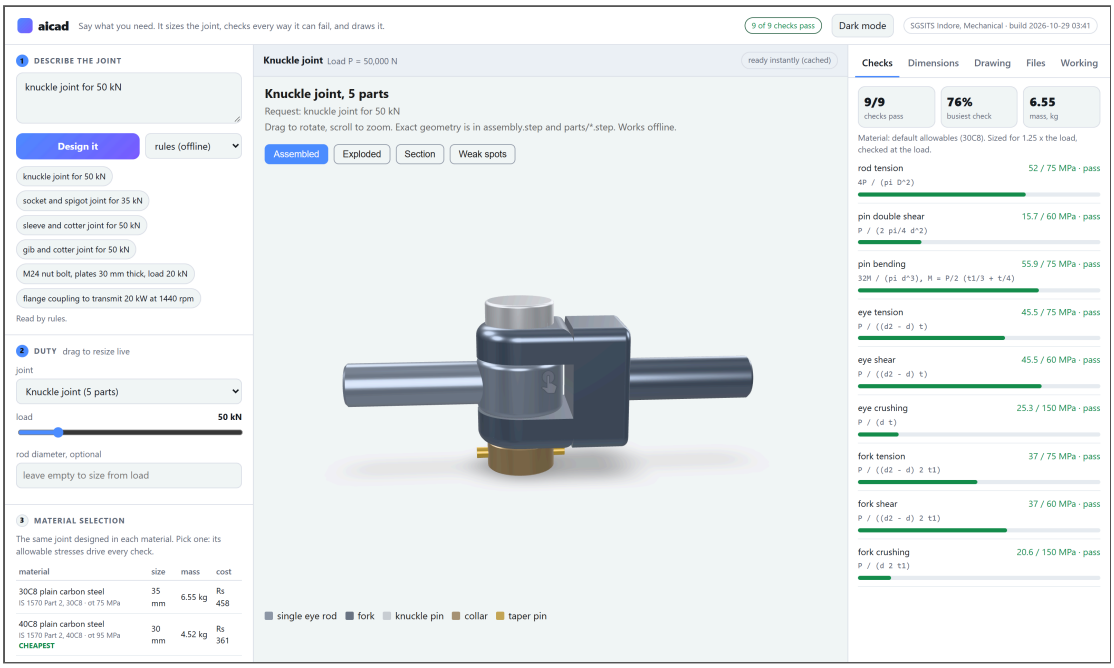


Figure A.1: Parameters extracted from the requirement, presented for confirmation or correction before the design proceeds

Checks | **Dimensions** | Drawing | Files | Working

D rod diameter rod in tension: D = sqrt(4P / (pi sigma_t)), rounded up to a preferred size	**35 mm**
d pin diameter governed by pin bending	**45 mm**
d2 eye outer diameter d2 = 2.0 x D, the rod diameter (outer diameter of eye), rounded up to a whole mm [NPTEL IIT KGP M4L2 fig 4.2.3.1]	**70 mm**
d3 pin head and collar diameter d3 = 1.5 x D, the rod diameter (pin head and collar diameter), rounded up to a preferred size [NPTEL IIT KGP M4L2 fig 4.2.3.1]	**55 mm**
t eye thickness t = 1.25 x D, the rod diameter (thickness of single eye), rounded up to a whole mm [NPTEL IIT KGP M4L2 fig 4.2.3.1]	**44 mm**
t1 fork plate thickness t1 = 0.75 x D, the rod diameter (thickness of each fork plate), rounded up to a whole mm [NPTEL IIT KGP M4L2 fig 4.2.3.1]	**27 mm**
t2 pin head thickness t2 = 0.5 x D, the rod diameter (thickness of pin head), rounded up to a whole mm [NPTEL IIT KGP M4L2 fig 4.2.3.1]	**18 mm**
collar_t collar thickness	**23 mm**
taper_d taper pin diameter (big end)	**9 mm**
clearance gap between eye and fork base	**5 mm**

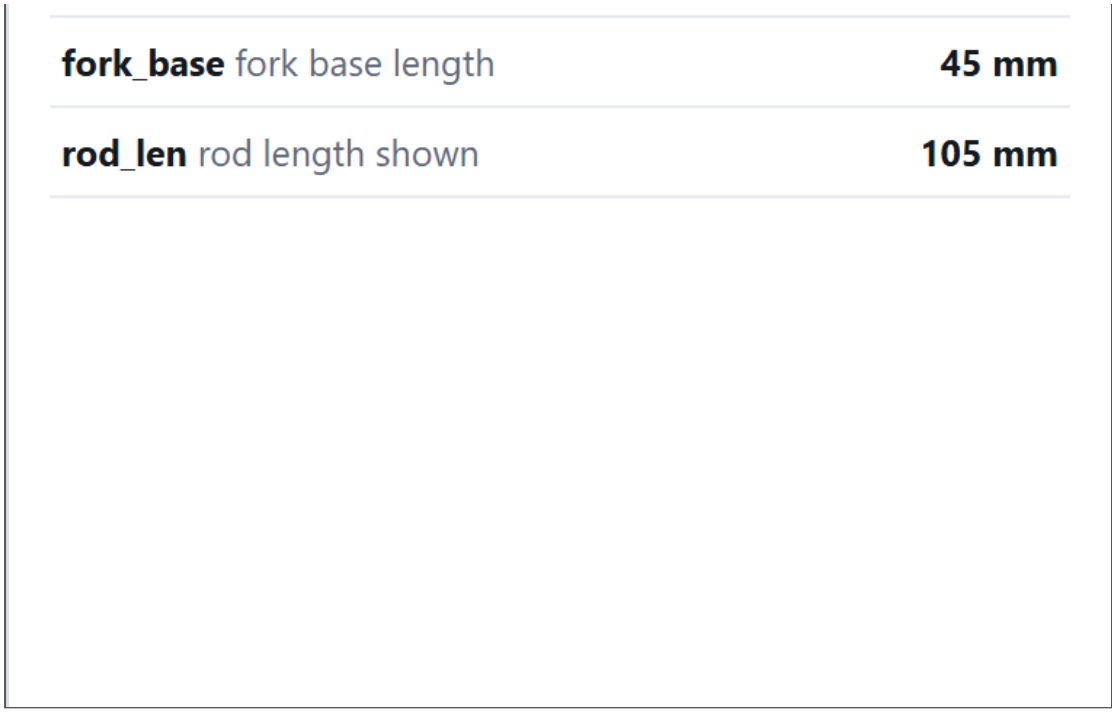


Figure A.2: Every computed dimension with the rule that produced it and its source

Checks | Dimensions | Drawing | Files | Working

9/9	76%	6.55
checks pass	busiest check	mass, kg

Material: default allowables (30C8). Sized for 1.25 x the load, checked at the load.

rod tension — 52 / 75 MPa · pass

`4P / (pi D^2)`

pin double shear — 15.7 / 60 MPa · pass

`P / (2 pi/4 d^2)`

pin bending — 55.9 / 75 MPa · pass

`32M / (pi d^3), M = P/2 (t1/3 + t/4)`

eye tension — 45.5 / 75 MPa · pass

`P / ((d2 - d) t)`

eye shear — 45.5 / 60 MPa · pass

`P / ((d2 - d) t)`

eye crushing — 25.3 / 150 MPa · pass

`P / (d t)`

fork tension — 37 / 75 MPa · pass

`P / ((d2 - d) 2 t1)`

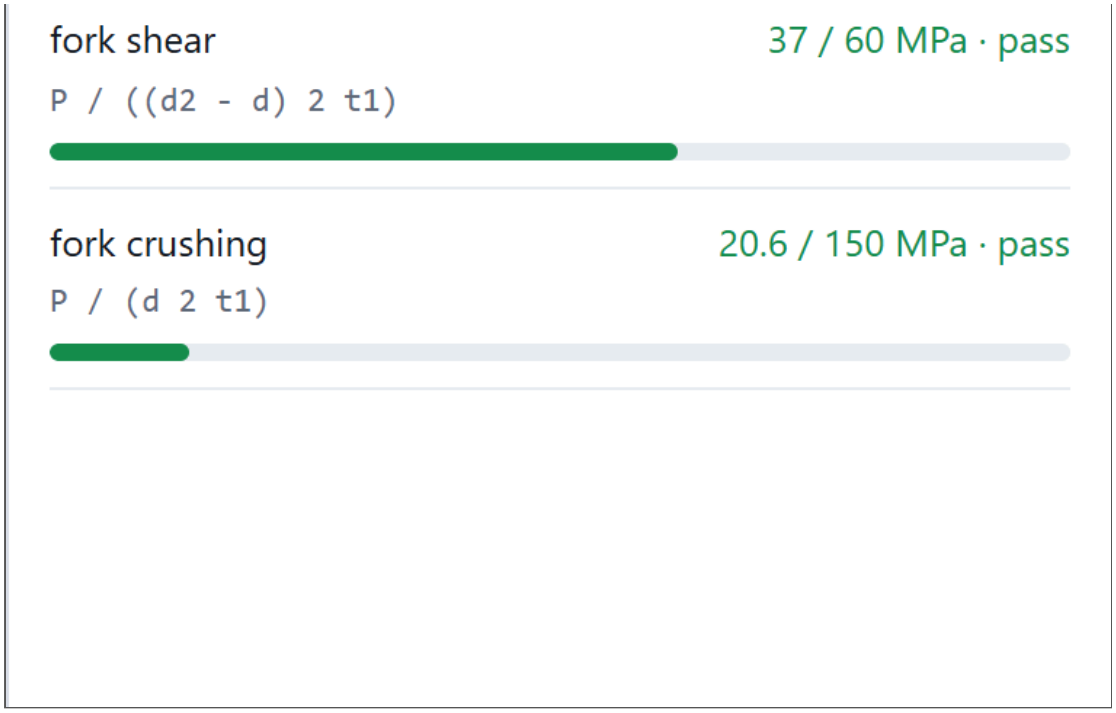
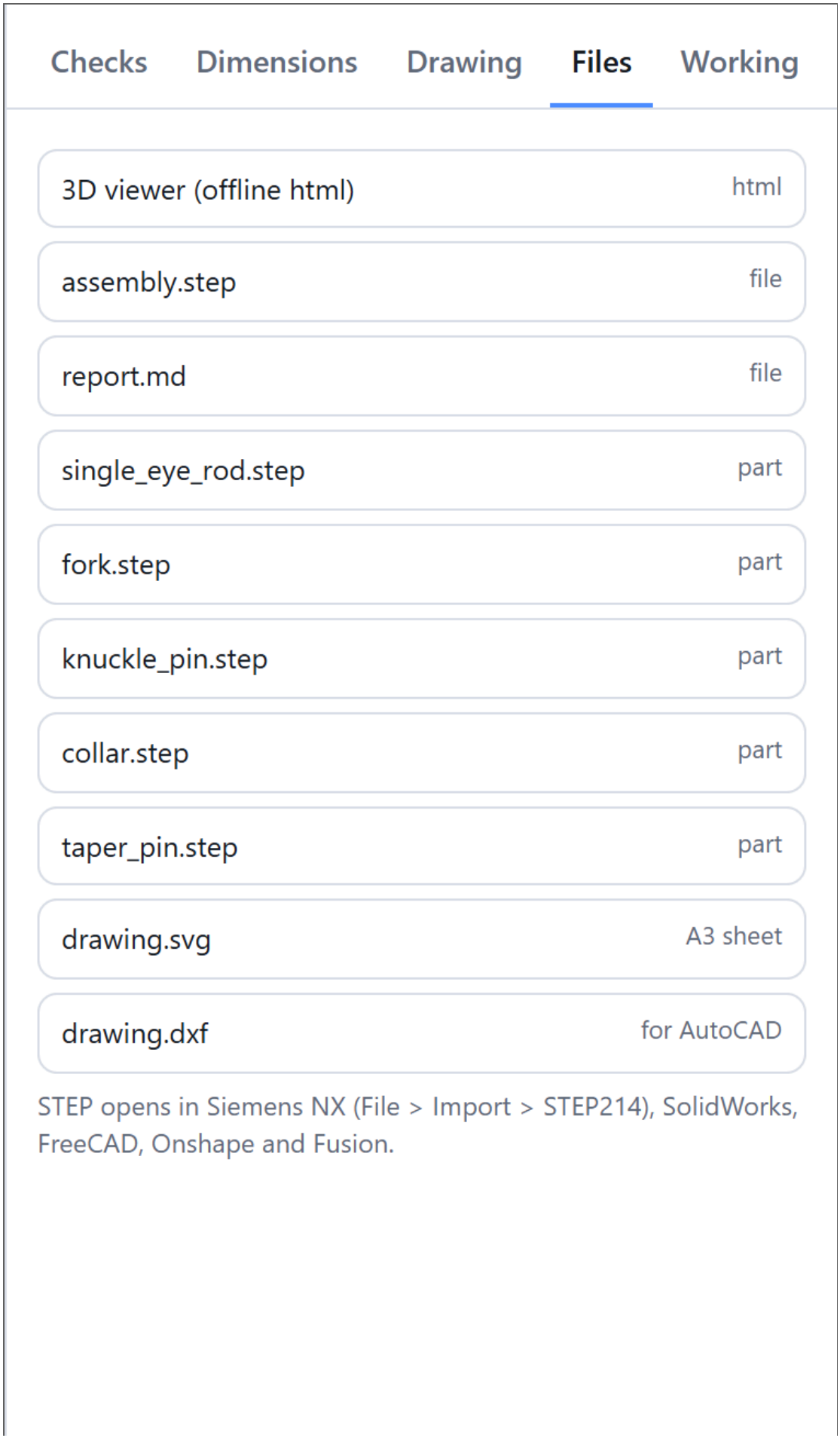


Figure A.3: Every failure check with its substituted arithmetic, its allowable stress and its utilisation


Checks
Dimensions
Drawing
Files
Working
3D viewer (offline html)
html
assembly.step
file
report.md
file
single_eye_rod.step
part
fork.step
part
knuckle_pin.step
part
collar.step
part
taper_pin.step
part
drawing.svg
A3 sheet
drawing.dxf
for AutoCAD
STEP opens in Siemens NX (File > Import > STEP214), SolidWorks, FreeCAD, Onshape and Fusion.

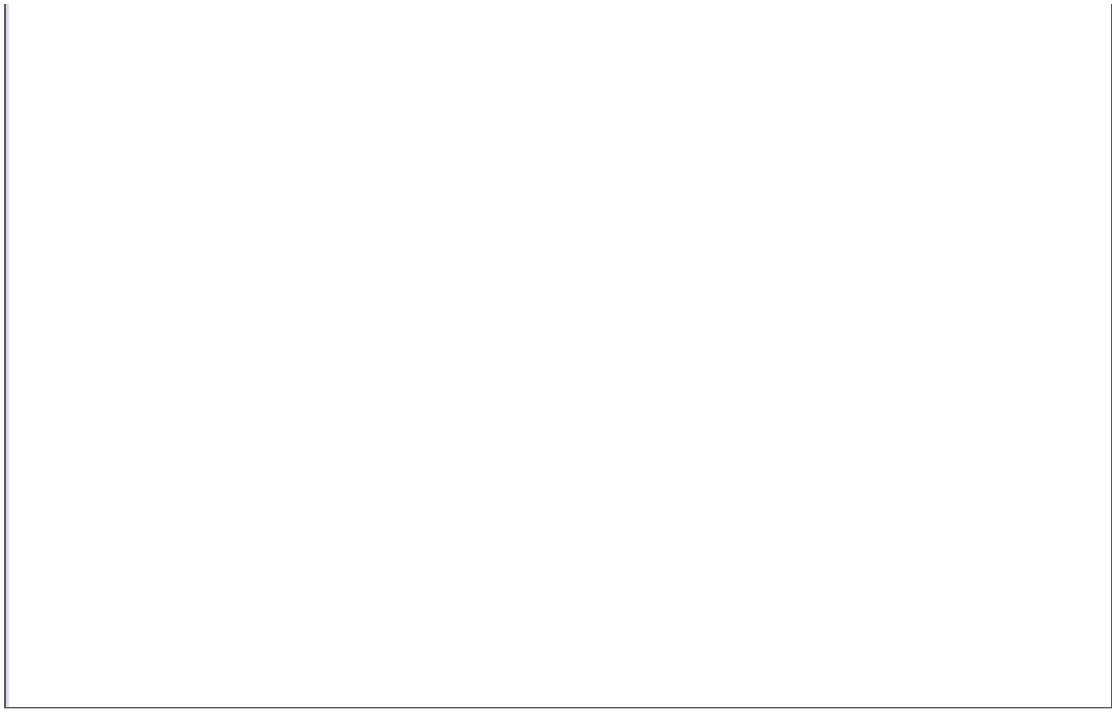


Figure A.4: Files produced by a single run, available for download

3 MATERIAL SELECTION

The same joint designed in each material. Pick one: its allowable stresses drive every check.

material	size	mass	cost
30C8 plain carbon steel IS 1570 Part 2, 30C8 · σt 75 MPa	35 mm	6.55 kg	Rs 458
40C8 plain carbon steel IS 1570 Part 2, 40C8 · σt 95 MPa **CHEAPEST**	30 mm	4.52 kg	Rs 361
45C8 plain carbon steel IS 1570 Part 2, 45C8 · σt 110 MPa	30 mm	4.22 kg	Rs 380
40Cr1 alloy steel IS 1570 Part 4, 40Cr1 · σt 160 MPa	25 mm	2.70 kg	Rs 405
Grey cast iron FG 260 IS 210, FG 260 · σt 40 MPa	45 mm	13.18 kg	Rs 791
Aluminium alloy 64430 (6061) IS 733, 64430 · σt 80 MPa **LIGHTEST**	35 mm	2.30 kg	Rs 643

Size is the governing diameter. Mass is the joint without the rod stubs. Allowable stresses and rates are working values, to be confirmed against the design data book and a supplier.

Figure A.5: Material selection: the joint designed in each material, with the governing size, mass and material cost

APPENDIX B

SELECTED SOURCE CODE

The listings below are taken unchanged from the source of the software at the time this report was generated. The complete source is available at github.com/Atishyy27/aicad.

Listing B.1: Rounding to a preferred size and the design factor (aicad/design.py)

```
def next_preferred(x: float, sizes: list[float]) -> float:
    for size in sizes:
        if size >= x - 1e-9:
            return float(size)
     raise ValueError(f"needs {x:.1f} mm, larger than the biggest size in the
table ({sizes[-1]} mm)")

def ceil_mm(x: float) -> float:
    return float(math.ceil(x - 1e-9))

def sizing_load(value, design_factor):
     """Size for a bigger load than asked for, then report the stresses at the
real load.
     A factor of 1.25 leaves the governing check at about 80 % of the allowed
stress."""
    if value is None:
        return None
    return value * max(1.0, float(design_factor or 1.0))
```

Listing B.2: Sizing the knuckle joint from the load (aicad/joints/knuckle.py)

```
def design(
    load_n=None,
    rod_diameter_mm=None,
    tension_mpa=None,
    shear_mpa=None,
    crushing_mpa=None,
    bending_mode=None,
    design_factor=None,
    **_ignored,
) -> Design:
    if load_n is None and rod_diameter_mm is None:
        raise ValueError("give a load or a rod diameter")
    check_positive(load_n, rod_diameter=rod_diameter_mm)
    s = stresses_from(DATA, tension_mpa, shear_mpa, crushing_mpa)
    sizes = DATA["preferred_diameters_mm"]
    props = DATA["proportions"]
      k_t, k_t1 = props["eye_thickness_t"]["times"], props["fork_thickness_t1"]
["times"]
    bending_mode = bending_mode or DATA["pin_bending"]["mode"]
         factor = float(design_factor if design_factor is not None else
DEFAULT_DESIGN_FACTOR)
    P = load_n
    Pd = sizing_load(P, factor)
    raw, notes = {}, []

    if rod_diameter_mm is not None:
        D = float(rod_diameter_mm)
    else:
        raw["rod_tension"] = math.sqrt(4 * Pd / (math.pi * s["tension"]))
        D = next_preferred(raw["rod_tension"], sizes)

     # the eye and fork thicknesses come from the ROD diameter (b = 1.25D, a =
0.75D) and are rounded
    # up to the next standard size before anything is sized off them, as Bhandari
does in Ex 4.5
    # (37.5 -> 40, 22.5 -> 25); the pin is sized separately and raised when shear
or bending needs it
```

```
      t_eye, t_fork = next_preferred(k_t * D, sizes), next_preferred(k_t1 * D,
sizes)
    d = props["pin_diameter_d"]["times"] * D
    governing = "pin = rod diameter"
    if P is not None:
        raw["pin_shear"] = math.sqrt(2 * Pd / (math.pi * s["shear"]))
        if raw["pin_shear"] > d + 1e-9:
             d, governing = next_preferred(raw["pin_shear"], sizes), "pin double
shear"
         # sigma_b = 16 P (t1/3 + t/4) / (pi d^3), with t and t1 fixed by the rod
diameter
          raw["pin_bending"] = (16 * Pd * (t_fork / 3 + t_eye / 4) / (math.pi *
s["bending"])) ** (1 / 3)
        if bending_mode == "size" and raw["pin_bending"] > d + 1e-9:
                 d, governing = next_preferred(raw["pin_bending"], sizes), "pin
bending"
            notes.append(
                                f"Pin raised to {d:g} mm for bending (needs
{raw['pin_bending']:.1f} mm), the same step "
                       "Bhandari Ex 4.5 and the NPTEL worked problem take. Set
pin_bending.mode to 'check_only' to keep the pin equal to the rod."
            )

     # the eye and fork outside diameter follows the PIN, d0 = 2d (Bhandari eq
4.26j), so a pin
    # raised for bending carries its eye with it. If a check still fails, grow it
further: the
    # NPTEL worked problem raises the pin but checks the eye against the old one,
and fails.
    d2 = props["eye_outer_diameter_d2"]["times"] * d
    if P is not None:
        grown = False
        for _ in range(400):
            trial = dict(D=D, d=d, d2=d2, t=t_eye, t1=t_fork)
                        if all(c.passed for c in _checks(Pd, trial, s) if
c.name.startswith(("eye", "fork"))):
                break
            d2 += 1
            grown = True
        if grown:
            notes.append(f"Eye and fork outer diameter grown to {d2:g} mm because
the bigger pin "
                         "leaves less metal around the hole.")

    dims = {
        "D": D,
        "d": d,
        "d2": d2,
             "d3": next_preferred(props["head_collar_diameter_d3"]["times"] * d,
sizes),
        "D1": next_preferred(props["enlarged_rod_D1"]["times"] * D, sizes),
        "t": t_eye,
        "t1": t_fork,
        "t2": ceil_mm(props["head_thickness_t2"]["times"] * D),
    }
    m = DATA["modelling_choices"]
    dims["collar_t"] = ceil_mm(times(m["collar_thickness"], d))
    dims["taper_d"] = ceil_mm(times(m["taper_pin_diameter"], D))
    dims["clearance"] = float(m["fork_clearance"]["mm"])
    dims["fork_base"] = times(m["fork_base_length"], d)
    dims["rod_len"] = m["rod_length"]["times"] * dims["d2"]
    dims = {k: round(v, 3) for k, v in dims.items()}

    def rule(key, prop_key, rounding="rounded up to a standard size"):
        p = props[prop_key]
        base = "d, the pin diameter" if p["of"] == "d" else "D, the rod diameter"
            return f"{key} = {p['times']} x {base} ({p['meaning']}), {rounding}
[{p['source']}]"

    rules = {
        "D": "given by user" if rod_diameter_mm is not None
```

```
           else "rod in tension: D = sqrt(4P / (pi sigma_t)), rounded up to a
preferred size",
        "d": f"governed by {governing}",
        "d2": rule("d2", "eye_outer_diameter_d2"),
        "d3": rule("d3", "head_collar_diameter_d3"),
        "D1": rule("D1", "enlarged_rod_D1"),
        "t": rule("t", "eye_thickness_t"),
        "t1": rule("t1", "fork_thickness_t1"),
        "t2": rule("t2", "head_thickness_t2", "rounded up to a whole mm"),
    }

    checks = _checks(P, dims, s) if P is not None else []

    if factor > 1 and P is not None:
         notes.insert(0, f"Sized for {factor:g} x the given load ({Pd:,.0f} N),
then checked at {P:,.0f} N. "
                         "That margin is why the checks sit below their limits.")
    return Design(
        joint=NAME, load_n=P, stresses=s, dims=dims, rules=rules, checks=checks,
notes=notes,
                                                          design_factor=factor,
verified_against_book=DATA["verified_against_book"], raw=raw,
    )
```

Listing B.3: The nine failure checks of the knuckle joint (`aicad/joints/knuckle.py`)

```
def _checks(P, dims, s):
    D, dd, d2, t, t1 = dims["D"], dims["d"], dims["d2"], dims["t"], dims["t1"]
    M = P / 2 * (t1 / 3 + t / 4)
    return [
         Check("rod tension", 4 * P / (math.pi * D**2), s["tension"], "4P / (pi
D^2)"),
        Check("pin double shear", P / (2 * math.pi / 4 * dd**2), s["shear"], "P /
(2 pi/4 d^2)"),
        Check("pin bending", 32 * M / (math.pi * dd**3), s["bending"], "32M / (pi
d^3), M = P/2 (t1/3 + t/4)"),
         Check("eye tension", P / ((d2 - dd) * t), s["tension"], "P / ((d2 - d)
t)"),
        Check("eye shear", P / ((d2 - dd) * t), s["shear"], "P / ((d2 - d) t)"),
        Check("eye crushing", P / (dd * t), s["crushing"], "P / (d t)"),
        Check("fork tension", P / ((d2 - dd) * 2 * t1), s["tension"], "P / ((d2 -
d) 2 t1)"),
         Check("fork shear", P / ((d2 - dd) * 2 * t1), s["shear"], "P / ((d2 - d)
2 t1)"),
        Check("fork crushing", P / (dd * 2 * t1), s["crushing"], "P / (d 2 t1)"),
    ]
```

Listing B.4: Safeguard: a number from the language model is used only if it appears in the requirement (`aicad/parse.py`)

```
def _numbers_in(text: str) -> list[float]:
    nums = [float(x) for x in re.findall(r"\d+(?:\.\d+)?", text)]
      nums += [v for w, v in _NUMBER_WORDS.items() if re.search(rf"\b{w}\b",
text.lower())]
    return nums

def _traceable(value: float, text: str, factors=(1,)) -> bool:
    for n in _numbers_in(text):
        for f in factors:
            if abs(n * f - value) <= 0.01 * max(abs(value), 1e-9):
                return True
    return False
```

Listing B.5: Material selection: the joint designed in every material (aicad/optimise.py)

```
def compare(spec, extra=None) -> list[dict]:
     """Every material in the table, plus any the user adds (same keys: name,
tension, shear,
    crushing, density, rate_inr_per_kg)."""
    joint = REGISTRY[spec.joint]
    rows = []
    for m in MATERIALS["materials"] + list(extra or []):
              s  =  replace(spec,  tension_mpa=m["tension"],  shear_mpa=m["shear"],
crushing_mpa=m["crushing"], notes=[])
        try:
            d = joint.design(**s.design_kwargs())
        except ValueError as exc:
            rows.append({"material": m["name"], "ok": False, "reason": str(exc)})
            continue
        vol = _volume_mm3(joint, d.dims)
        mass = vol * 1e-9 * m["density"]
        rows.append({
                 "material": m["name"], "is_grade": m.get("is_grade", ""), "ok":
d.all_passed, "mass_kg": mass,
                             "cost_inr":  mass  *  m["rate_inr_per_kg"],  "rate":
m["rate_inr_per_kg"],
            "size": d.dims.get("D", d.dims.get("d")), "spec": s,
        })
    return rows
```

Listing B.6: Faces cut by the plane of section A-A, for hatching (aicad/freecad_pack.py)

```
def section_loops(joint, dims) -> list:
     """Outlines of the faces cut by the plane y = 0, part by part, as 3D point
loops.
     Computed here with CadQuery, where it can be tested, and only drawn inside
FreeCAD."""
    out = []
    try:
        import cadquery as cq
        from .joints._shapes import PLAIN_EDGES
        PLAIN_EDGES.on = False
        for name, wp in joint.build_parts(dims).items():
            if any(w in name for w in UNSECTIONED):
                continue
            solid = wp.val()
            bb = solid.BoundingBox()
            big = 4 * max(bb.xlen, bb.ylen, bb.zlen, 1.0)
             half = solid.cut(cq.Solid.makeBox(big, big, big, cq.Vector(-big / 2,
-big, -big / 2)))
            loops = []
            for f in half.Faces():
                fb = f.BoundingBox()
                if fb.ylen > 1e-3 or abs(fb.ymin) > 1e-3:
                       continue                          # not a face lying in the
cutting plane
                for w in [f.outerWire()] + f.innerWires():
                           pts = [(round(v.x, 3), 0.0, round(v.z, 3)) for v in
w.positions([i / 96 for i in range(96)])]
                    loops.append(pts)
            if loops:
                out.append({"part": name, "loops": loops})
    except Exception:
        return []
    return out
```

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